

Development Plan

ProgName

Team #, Team Name
Hemal Rajput
Gurmanjot Minhas
James Araiche
Lily Porter
Sarah Dorfman

Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

[Put your introductory blurb here. Often the blurb is a brief roadmap of what is contained in the report. —SS]

[Additional information on the development plan can be found in the [lecture slides](#). —SS]

1 Confidential Information?

[State whether your project has confidential information from industry, or not. If there is confidential information, point to the agreement you have in place. —SS]

[For most teams this section will just state that there is no confidential information to protect. —SS]

2 IP to Protect

[State whether there is IP to protect. If there is, point to the agreement. All students who are working on a project that requires an IP agreement are also required to sign the “Intellectual Property Guide Acknowledgement.” —SS]

3 Copyright License

Our team is adopting the GNU General Public License (GPL). The license is available in the [LICENSE file](#) at the root of our repository.

4 Team Meeting Plan

In-person team meetings will be held weekly on Thursdays from 3:30pm to 4:20pm. The purpose of these meetings will be to discuss upcoming deliverables, progress updates, and potential blockers. They will also be used to consolidate any questions that team members may have for the instructional team and project supervisors.

There will also be biweekly meetings held with the project supervisors on Fridays from 9:00am to 10:00am. They are planned to be in-person but may be shifted to virtual Microsoft Teams meetings based on availability. These will be used to ensure the project is in line with the advisors’ requirements, and to address any questions that the development team might have.

The meeting chair (Sarah Dorfman) will prepare an agenda for the main topics of discussion and distribute it to the team at least one day prior to any team meetings to give all members time to prepare. All members are encouraged to take notes during the meeting. After the meeting, the team will collectively assign action items to each other. If there is a supervisor meeting planned for that week, the team will also plan a corresponding agenda.

5 Team Communication Plan

[Issues on GitHub should be part of your communication plan. —SS]

6 Team Member Roles

To ensure the team works collaboratively and effectively, we have defined unique roles for each member. Each role has its own clearly defined tasks and responsibilities that reflect the team's goals and individual strengths of team members. While each team member will have a primary area of focus, these roles are not fixed and may evolve as the project progresses and team members rotate responsibilities:

- **Project Manager & CI/CD/Admin Lead:** This role oversees and manages the project timeline, team organization and progress. The project manager will maintain the project's CI/CD pipeline to ensure LaTeX documentation compiles and deploys correctly. They will also manage repository permissions, create issues, track team progress, and organize milestones and deadlines.
- **Team Liaison:** This role is the main point-of-contact between the team and course instructors. They are responsible for submitting/creating pull requests for deliverables, and facilitating communication of feedback and comments between the instructor and the group.
- **Requirements Expert:** This role leads communication with project stakeholders. They will lead the gathering of system and stakeholder goals and lead the process of translating them into clear requirements to support the writing of the Software Requirements Specification (SRS) documentation, ensuring traceability between stakeholder goals, system requirements and the final design.
- **Systems Architect/Development Lead:** This role will ensure that the high-level system architecture is followed and oversees the core implementation of the software. They will ensure the team adheres to the defined architecture and design principles and ensure that the system is designed with extensibility to accommodate changes and project expansion.
- **Machine Learning (ML) Lead:** This role is responsible for overseeing the design, implementation, and evaluation of the project's machine learning components. They will be the primary manager/liaison for datasets, model training, and performance evaluation.
- **User Interface & Experience (UI/UX) Lead:** This role is responsible for overseeing the design and implementation of the user interface. They will be the primary manager/liaison for the creation of wireframes, prototypes and the project's adherence to human-centred design.

7 Workflow Plan

- How will you be using git, including branches, pull request, etc.?
- How will you be managing issues, including template issues, issue classification, etc.?
- Use of CI/CD

8 Project Decomposition and Scheduling

- How will you be using GitHub projects?
- Include a link to your GitHub project

[How will the project be scheduled? This is the big picture schedule, not details. You will need to reproduce information that is in the course outline for deadlines. —SS]

9 Proof of Concept Demonstration Plan

At a high level, this application will listen to child speech, process it into its phonetic representation, and analyze this to give feedback to SLPs. The greatest risks associated with this lie in the intermediate audio processing stage:

1. **Converting Child Speech To Phonetic Representation** - Since children are in a developmental phase, their speech patterns vary significantly when compared to adults. Therefore, it may be challenging to ensure an acceptable accuracy rate when converting child speech into phonetic representations.
2. **Ensuring An Acceptable Processing Time** - For this application to be viable in a clinical setting, it will need to give real time feedback to the SLP during an appointment on their computers. As per the project supervisors, existing child speech interpretation models are quite slow, so it may be difficult for this one to process the data fast enough to give the clinician feedback in less time than it would have taken for them to reach a deduction themselves.

To ensure these risks can be overcome, a suitable training dataset must be found online. To ensure speech to phonetic conversion accuracy, there must be sufficient data to properly train the model. It is important to note that this data must be publicly available to comply with HIPAA patient confidentiality guidelines.

Once this data is sourced, we will need to develop the actual model that takes child speech and converts it into a readable format with the corresponding

phoneme representations. If this model meets the established accuracy benchmarks, it will be feasible to continue designing this application for pediatric speech purposes. Moreover, it is fair to assume that speech therapy offices will use typical office computers that are not equipped with powerful hardware. Therefore, it is necessary to show that the model can meet the benchmark processing time on a device of similar specifications.

10 Expected Technology

11 Expected Technology

11.1 Languages and Constraints

- **Python:** speech processing, model evaluation and training, and application backend.
- **JavaScript:** user interface.
- **SQL:** data storage and access control.
- **Git, GitHub, and GitHub Projects:** course requirements and will be used for version control, issue tracking, code review, and planning.
- The application is web-based so it will run on devices that a typical clinic would have, i.e. a desktop.

11.2 Libraries and Frameworks

- **Backend:** an Python web framework (e.g., FastAPI) with data validation (e.g., Pydantic).
- **Frontend:** a JavaScript framework (e.g., React), an API to capture audio from the browser.
- **Data:** a relational database (e.g., PostgreSQL) with row-level access control, hosted or self-hosted.
- **Speech and machine learning:** PyTorch, the Hugging Face Transformers library, and standard audio-processing libraries.

11.3 Pre-trained Models and Training

- We expect to start from pre-trained speech models (e.g., wav2vec2) and existing phone recognizers, to train a model, if the existing solutions are found to not be good enough.
- Training data will be publicly available child-speech corpora with phonetic annotations

- Both existing commercial and open-source models are evaluated and are a fallback if a trained model does not meet the agreed targets

Components we expect to implement ourselves.

- comparison of recognized phones against the target sound
- an evaluation harness for accuracy, speed, and regression testing
- the application: practice interface, curriculum, and clinician dashboard.

11.4 Development Tooling

- **Linting and formatting:** Ruff for Python; Prettier for JavaScript.
- **Unit testing:** pytest for Python; Vitest for JavaScript.
- **Continuous Integration:** Pull requests must have a reviewer; branch protection blocking tests that fail; unit testing to prevent breaking the application on code that compiles on merge;

11.5 Performance Measurement

- **Accuracy:** phone error rate against reference transcriptions; precision and recall of substitution detection per target sound; agreement with speech-language pathologist annotations.
- **Speed:** real-time factor (processing time divided by audio duration) and end-to-end latency, measure at each model's size to map the accuracy-versus-latency tradeoff.
- **Regression testing:** a fixed held-out evaluation set is re-run after every training run or model change, and a change is accepted only if results stay within agreed thresholds.

11.6 Areas Requiring Additional Training

- Phonetics and speech sound disorders, with our SLP advisor.
- Fine-tuning and evaluating speech models.
- Database access-control design.

12 Use of AI and LLMs

Our team will use LLMs to support brainstorming, application development, and documentation. Tools such as Claude and ChatGPT may be used to explore design approaches, explain concepts, and write code. In the coding process, agents will also be implemented to improve workflow and follow best practices. Team members will make and take responsibility for all project decisions and submitted work.

We will review any generated code before including it in the application. We will check that it meets the project requirements, follows our design, and works as intended by running relevant builds, checks, and tests. Any output that cannot be verified or does not meet these checks will be revised or discarded.

13 Coding Standard

[What coding standard will you adopt? —SS]

14 Appendix — Generative AI Use Disclosure

All writing was done by the team members by hand. All research and facts were obtained through academic sources, or by consulting the project supervisors unless explicitly stated below.

14.1 AI Tools Used

Table 2: List of AI Tools Used

Tool Number	Name	Model
1	Claude	Sonnet 5
2	ChatGPT	GPT 5 Thinking Mini

14.2 How and Where Used

Table 3: Usage Details for All AI Tools Used

Tool Number	Location	Details
1	Bibliography	Used to insert a reference to the references.bib file.
2	Problem Statement and Goals; (1.3) Stakeholders	Instructed the AI to think from the perspective of a patient’s parent when assessing stakeholders: highlighted privacy concerns as another consideration for this stakeholder.

14.3 Verification of AI-Generated Content

As none of the AI usage was for factual content, verification was very simple. For tool 1, we simply verified the resulting citation for correctness in the final output. For tool 2, it was used as more of a brainstorming utility, so only unconsidered elements that made logical sense were included.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. Why is it important to create a development plan prior to starting the project?

Much of the lost time in any project, especially a group project, is due to disorganization and poor coordination. Developing that organizational framework ensures that the team is on track to complete the project in a timely manner. The value of development planning rather than jumping straight into the code, is ensuring that more of the thinking is done sooner. The later work is instead coordinated implementation. Communication is one of the most challenging parts of engineering. We can ensure that we all know our expectations of the project, and we can resolve miscommunication early on.

2. In your opinion, what are the advantages and disadvantages of using CI/CD?

CI/CD automates much of the menial labour in software development. This lets the team focus on more critical aspects of the work instead of getting bogged down by the friction of testing and deployment. It is also a significant time saver. Many mistakes, which may be missed by human reviewers, will be caught by the CI/CD pipeline. This means reviewers can focus on architectural and design details as opposed to spending time on trivial things, such as naming conventions.

On the other hand, the main disadvantage of implementing CI/CD is the initial time spent setting it up. This is especially true for us, since none of us have implemented CI/CD in our own projects, even though we may have worked in codebases that had CI/CD pipelines during our co-ops. Spending too much time on the perfect CI/CD pipeline is time lost on the features of the project.

3. What disagreements did your group have in this deliverable, if any, and how did you resolve them?

We had some disagreements on the Expected Technology section, specifically about the use of Cloud-based and web solutions for our project. This project will likely require storing and analyzing the user's medical data, which means we need to consider the ethics and security of our solution. Any web-based application would require considering cybersecurity, and any Cloud-based databases would require detailed inspections of the provider's security and privacy policies. However, these approaches would significantly ease the technical development of our project, allowing us to focus on meeting our goals. We had to spend some time considering the trade-offs of each approach and come to a compromise to move on to other aspects.

Appendix — Team Charter

[borrows from [University of Portland Team Charter](#) —SS]

External Goals

[What are your team’s external goals for this project? These are not the goals related to the functionality or quality of the project. These are the goals on what the team wishes to achieve with the project. Potential goals are to win a prize at the Capstone EXPO, or to have something to talk about in interviews, or to get an A+, etc. —SS]

Attendance

Expectations

[What are your team’s expectations regarding meeting attendance (being on time, leaving early, missing meetings, etc.)? —SS]

Acceptable Excuse

[What constitutes an acceptable excuse for missing a meeting or a deadline? What types of excuses will not be considered acceptable? —SS]

In Case of Emergency

[What process will team members follow if they have an emergency and cannot attend a team meeting or complete their individual work promised for a team deliverable? —SS]

Accountability and Teamwork

Quality

[What are your team’s expectations regarding the quality of team members’ preparation for team meetings and the quality of the deliverables that members bring to the team? —SS]

Attitude

[What are your team’s expectations regarding team members’ ideas, interactions with the team, cooperation, attitudes, and anything else regarding team member contributions? Do you want to introduce a code of conduct? Do you want a conflict resolution plan? Can adopt existing codes of conduct. —SS]

Stay on Track

[What methods will be used to keep the team on track? How will your team ensure that members contribute as expected to the team and that the team performs as expected? How will your team reward members who do well and manage members whose performance is below expectations? What are the consequences for someone not contributing their fair share? —SS]

[You may wish to use the project management metrics collected for the TA and instructor for this. —SS]

[You can set target metrics for attendance, commits, etc. What are the consequences if someone doesn't hit their targets? Do they need to bring the coffee to the next team meeting? Does the team need to make an appointment with their TA, or the instructor? Are there incentives for reaching targets early? —SS]

Team Building

[How will you build team cohesion (fun time, group rituals, etc.)? —SS]

Decision Making

[How will you make decisions in your group? Consensus? Vote? How will you handle disagreements? —SS]